

World Model Presentation

Submitted To

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1 Semantic World Model

World Models are a type of model capable of predicting the future given a large amount of data.

A generalized world model that is represented as an action-conditional vision-language model that answers questions about the semantic effects of actions in the future.

These types of models utilise Multimodal Data, which leads to the question, “What is multimodal data”?

Multimodal data is a combination of different types of data, such as text, images, audio, video, and sensor readings, that are combined and used in a single system to complete complex tasks, such as predicting the next action.

In this paper titled, **Semantic World Models**, Jacob et al. introduce a generalized world model that is represented as an action-conditional vision-language model that answers questions about frames, more accurately future frames given the current observations and a sequence of actions.

SWM was evaluated on a set of multiple different tasks in two commonly used multi-task simulation domains – Language Table (LangTable) and OGBench.

These evaluations show two observations:

1. SWM can accurately answer questions about future outcomes while generalizing to novel scenes.
2. SWM can be combined with standard sampling-based planning techniques and a gradient-based improvement technique to solve diverse robotic tasks with considerable policy improvement through test-time optimization.

1.1 Dataset

To train this SWM, Jacob et al. generated a SAQA dataset, which is a state-action-question-answer dataset.

$$D_{SAQA} = \{(S_i, a_{i:j}, Q_{S_j}, A_{S_j}) \dots\}$$

where

$$j = i + h$$

Here:

S_i = Current state (RGB frame in our case)

h = horizon

$a_{i:j}$ = Sequence of actions taken from state S_i

Q_{S_j}, A_{S_j} = Question-answer tuple about future state S_j

- **Horizon (h):** The total number of time steps (clock ticks) to get from the starting state S_i to the future state S_j .
- **Action Sequence ($a_{i:j}$):** The ordered list of actions the agent takes during those h steps to bridge the gap.

1.2 Architecture

The SWM architecture is based on an open-source VLM (Video Language Model) named PaliGemma.

SWM contains three core pretrained components:

1. A transformer-based autoregressive language model with token embedding size d_{tok} .
2. A vision encoder v_{Φ} with a feature size d_{img} .
3. A projection matrix

$$W \in \mathbb{R}^{d_{tok} \times d_{img}}$$

This paper uses a 3B parameter checkpoint from PaliGemma as the base model.

To adapt the base model (PaliGemma 3B Parameter Checkpoint) to answer questions about a specific future as a result of the actions, the model needs to be conditioned on the actions.

In a semantic world model, planning is a live decision-making process using high-level concepts. Here, they used two types of planning – Sampling-based planning and gradient-based planning. Another planning strategy was used for Multistep tasks, with the idea that SWM can be used to track task progress and transition between subgoals without requiring any additional components.

1.3 Results

SWM demonstrates an average performance increase over the base policies from 14.4% to 81.6% on average for LangTable and 45.33% to 76% on average for OFBench.

SWM also outperforms both AVD (Action Conditioned Video Diffusion) and IDQL (Implicit Diffusion Q-Learning) baselines across all tasks, demonstrating the effectiveness of SWM for planning.

2 Revisiting Feature Prediction for Learning Visual Representations from Video

In this paper, Adrien et al. use unsupervised learning from videos for feature prediction as a stand-alone objective. They introduce V-JEPA or Video Joint Embedding Predictive Architectures.

In essence, V-JEPA is trained with an objective of feature prediction, which is far superior to the models that either predict or reconstruct the pixels.

2.1 Architecture

Adrien et al. used the JEPA or Joint Embedding Predictive Architecture as the base model to train on videos.

The basic JEPA architecture is made up of an encoder, which computes the representation of the inputs, and a predictor, which predicts the representation of the target from the representation of the input, conditioned on a variable indicating the transformation between the input and the target.

Their training objective was to satisfy the constraint that representations computed from one part of the target video should be predictable from representations computed from another part of the input video.

To predict the target from the input, the authors used a masked modeling formulation. They leverage two types of masks:

- Union of 8 randomly sampled target blocks covering 15% of each frame.
- Union of 2 randomly sampled target blocks covering 70% of a frame.

The aspect ratio for all sampled blocks is randomly chosen in the range (0.75, 1.5).

The result is an average masking ratio of approximately 90%.

Adrien et al. use a Vision Transformer (ViT) to process a video.

2.2 Dataset

Adrien et al. combined several public datasets to construct an unsupervised video pre-training dataset referred to as **VideoMix2M**.

The dataset combines:

- HowTo100M (HT)
- Kinetics-400/600/700 (K710)
- Something-Something-v2 (SSv2)

resulting in approximately 2 million videos.

2.3 Evaluations

The authors evaluated the pre-trained models on downstream video and image tasks.

Video tasks included:

- Action recognition on Kinetics-400 (K400)
- Motion classification on Something-Something-v2 (SSv2)
- Action localization on AVA

All models are pretrained on VideoMix2M for 90K iterations with a batch size of 3072 using multi-block masking.

Performance was examined on:

- Kinetics-400 (K400)
- Something-Something-v2 (SSv2)
- ImageNet-1K (IN1K)

using a frozen backbone with an attentive probe and reporting top-1 accuracy using a single centre view.

They also examined end-to-end fine-tuning performance on Kinetics-400.

3 Video Representation Learning with Joint-Embedding Predictive Architecture

In this paper, Katrina et al. present Video JEPA with Variance-Covariance Regularization (VJ-VCR): a joint embedding predictive architecture for self-supervised video representation learning that employs variance and covariance regularization to avoid representation collapse.

Self-supervised learning for videos enables models to learn and extract high-level, information-rich features from videos without external annotations.

Generative models are used for pretext tasks where they make predictions in the input pixel space. JEPA is an alternative to these generative models.

In JEPA, prediction occurs in abstract space instead of pixel space, which is less computationally expensive.

There is a key challenge in training JEPA: preventing collapse in the hidden representations.

To prevent this collapse, Katrina et al. propose applying variance-covariance regularization to hidden representations.

3.1 Architecture

The VJ-VCR model has two main parts:

- Encoder
- Predictor

and it can also incorporate a latent variable.

The Encoder maps all input frames and target frames from the video.

The Predictor takes hidden states of the input frames and predicts the hidden states of the target frames.

VJ-VCR incorporates a latent variable to facilitate prediction when target frames are not completely deterministic versions of the input.

3.2 Experiments

To evaluate VJ-VCR, Katrina et al. performed the following experiments:

1. Generative Model Baseline comparisons.
2. Validation on two types of datasets:
 - Deterministic datasets: MovingMNIST and CLEVR
 - Non-deterministic datasets: Stochastic MovingMNIST and CATER

3.3 Results

Deterministic Setting

1. MovingMNIST: First two VJ-VCR models achieved MSE of 0.04. The third achieved MSE of 0.15.

2. CLEVR: RankMe score of 359.8 with decoder and 427.4 without decoder.

Non-deterministic Setting

1. MovingMNIST: Discrete latent variables predicted trajectory switches with 80% accuracy.
2. CATER: VJ-VCR outperformed generative models by 13.6%.

4 Self-Supervised Video Models Enable Understanding, Prediction and Planning

In this paper, Assran et al. explored a self-supervised approach that combines internet-scale video data with a small amount of interaction data (robot trajectories), to develop models capable of planning, predicting and acting in the physical world.

The authors leverage the Joint Embedding Predictive Architecture (JEPA), which learns by making predictions in a learned representation space.

For V-JEPA 2, they use a stage-wise training procedure consisting of action-free pre-training followed by post-training with interaction data.

The V-JEPA 2 encoder was trained with up to 1 billion parameters and more than 1 million hours of video.

After pretraining, they trained an action-conditioned world model called V-JEPA 2-AC.

This is a 300M-parameter transformer employing a block-causal attention mechanism.

4.1 Architecture

The Encoder and Predictor are parameterized as Vision Transformers (ViTs).

They leverage RoPE (Rotary Position Embedding) instead of absolute sincos positional encoding.

Key scaling ingredients include:

1. Data scaling from 2M to 22M videos.
2. Model scaling from 300M to over 1B parameters.
3. Longer training from 90K to 252K iterations.
4. Higher resolution video and longer clips.

4.2 Evaluation

Evaluation categories:

- Understanding
- Prediction
- Planning

Key results include:

- 77.3 top-1 accuracy on Something-Something v2.
- State-of-the-art Video Question Answering performance.
- 39.7 Recall@5 on Epic-Kitchens-100.
- Successful robotic manipulation planning with V-JEPA 2-AC.

5 Self-Supervised Learning from Images with a Joint-Embedding Predictive Architecture

In this paper, Assran et al. present I-JEPA (Image-based Joint Embedding Predictive Architecture), a non-generative approach for self-supervised learning from images.

The idea behind I-JEPA is simple: from a single context block, predict representations of various target blocks in the same image.

I-JEPA predicts missing information in an abstract representation space, avoiding unnecessary pixel-level details.

5.1 Architecture

I-JEPA uses the JEPA framework.

The context encoder is a Vision Transformer (ViT) that processes visible context patches.

The predictor is a narrow ViT conditioned on positional tokens to predict target block representations.

Target encoder weights are updated using exponential moving average.

5.2 Evaluation

- ViT-L/16 achieved Top-1 accuracy of 69.4 after 600 epochs.
- ViT-H/14 achieved 73.3 after 300 epochs.
- ViT-H/16₄₄₄ achieved 77.3 after 300 epochs.

Linear evaluation results:

- CIFAR10: 87.5
- Places205: 58.4
- iNet18: 47.6

Low-level vision tasks:

- CLEVR Object Counting: 86.7
- Depth Prediction: 72.4

6 LeWorldModel: Stable End-to-End Joint-Embedding Predictive Architecture from Pixels

In this paper, Lucas et al. introduce LeWorldModel (LeWM), which trains stably end-to-end from raw pixels using only two loss terms:

- Next-embedding prediction loss
- Gaussian latent regularization

Their experiments demonstrate competitive control performance across diverse 2D and 3D tasks while enabling planning up to 48x faster.

6.1 Architecture

LeWM consists of two components:

- Encoder
- Predictor

The encoder maps frame observations into latent representations:

$$z_t = enc_{\theta}(o_t)$$

The predictor models environment dynamics:

$$z_{t+1} = pred_{\phi}(z_t, a_t)$$

The encoder is implemented as a Vision Transformer (ViT).
The predictor is a transformer with:

- 6 layers
- 16 attention heads
- 10% dropout

Actions are incorporated using Adaptive Layer Normalization (AdaLN).

6.2 Dataset

LeWorldModel is trained in a fully offline and reward-free setting.

Datasets include:

- TwoRoom
- PushT
- OGBench-Cube
- Reacher

6.3 Results

LeWM achieved significantly better performance than PLDM while remaining competitive with DINO-WM.

- Agent Location:
 - Linear: 0.052 ± 0.149
 - MLP: 0.004 ± 0.056
- Block Location:
 - Linear: 0.029 ± 0.073
 - MLP: 0.001 ± 0.006
- Block Angle:
 - Linear: 0.187 ± 0.359
 - MLP: 0.021 ± 0.139

7 Video World Models with Long-term Spatial Memory

In this paper, Tong et al. introduce a novel framework for improving long-term consistency in video world models through a geometry-grounded long-term spatial memory. The authors argue that existing video generation models often struggle to maintain consistency over long temporal horizons due to limited memory of previously observed scenes.

To address this challenge, they propose a memory-augmented video world model that explicitly stores and utilizes 3D scene information during generation. Their framework incorporates three distinct forms of memory: spatial memory, working memory, and episodic memory, each represented through dedicated mechanisms. By combining these memory systems, the model is capable of generating videos with improved temporal coherence and scene consistency.

The framework relies on a set of recently generated context frames together with long-term memory representations to guide future frame prediction.

7.1 Architecture

The proposed system is a latent video generation model implemented using a Diffusion Transformer (DiT) and conditioned on three complementary memory mechanisms during autoregressive frame generation.

The three memory components are:

1. **Working Memory:** Recent context frames provide short-term information about ongoing scene dynamics.
2. **Spatial Memory:** A point cloud representation is autoregressively generated alongside the video frames. This geometry-grounded memory captures static scene structures and enables long-term spatial consistency.

3. **Episodic Memory:** A collection of historical reference frames is stored as sparse long-term memory, allowing the model to recall previously observed visual content.

Tong et al.’s prototype implementation is built upon CogVideoX, a state-of-the-art video generation framework based on a two-stage architecture.

CogVideoX employs a Diffusion Transformer with 3D attention blocks to model latent video representations. The proposed framework extends this architecture by introducing additional control signals derived from the three memory systems, enabling consistent long-term video generation.

7.2 Dataset

The authors construct a custom dataset from raw videos collected from MiraData.

Each video is segmented into multiple clips containing 97 frames. For every clip:

- The first 49 frames serve as the source sequence.
- The remaining 48 frames serve as the target sequence.
- A shared transition frame is used to preserve temporal continuity.

To recover scene geometry, the authors perform 4D reconstruction using Mega-SaM. This process extracts:

- Camera intrinsics
- Camera extrinsics
- Per-frame depth maps

The recovered RGB-D observations are then integrated into a volumetric representation using TSDF-Fusion, producing the spatial memory used during training and inference.

7.3 Evaluation

The proposed model is evaluated using several standard video quality and consistency metrics.

The reported results are:

- PSNR (Peak Signal-to-Noise Ratio): 19.10
- SSIM (Structural Similarity Index Measure): 0.6471
- LPIPS (Learned Perceptual Image Patch Similarity): 0.3069
- Camera Accuracy: 0.3626

The experimental results demonstrate that incorporating geometry-grounded long-term spatial memory significantly improves temporal consistency and scene coherence compared to conventional video world models that rely only on short-term context information.

8 VL-JEPA: Joint Embedding Predictive Architecture for Vision-Language

In this paper, Delong et al. introduce VL-JEPA, a vision-language model built upon the Joint Embedding Predictive Architecture (JEPA). Unlike traditional Vision-Language Models (VLMs) that generate text tokens autoregressively, VL-JEPA predicts continuous embeddings of target texts in a latent representation space.

The core idea is that learning in an abstract representation space allows the model to focus on task-relevant semantic information while abstracting away task-irrelevant linguistic details. This enables more efficient and robust vision-language learning.

The authors summarize their main contributions as follows:

1. They introduce VL-JEPA, the first non-generative model based on JEPA capable of performing general-domain vision-language tasks in real time.
2. They demonstrate through controlled experiments that latent-space embedding prediction outperforms data-space token prediction used in conventional VLMs.
3. They show that VL-JEPA achieves substantial efficiency improvements for online video-streaming applications due to its non-autoregressive architecture and support for selective decoding.
4. They demonstrate that VL-JEPA_{SFT} can effectively handle classification, retrieval, and visual question answering tasks within a unified model architecture.

8.1 Architecture

VL-JEPA consists of four main components:

1. **X-Encoder:** The X-Encoder processes visual inputs and compresses them into compact visual embeddings. These embeddings are continuous vector representations analogous to visual tokens used in conventional vision-language models.
2. **Predictor:** The Predictor is the central component of VL-JEPA. It maps visual embeddings to a prediction of the target embedding while being conditioned on a textual query.
3. **Y-Encoder:** The Y-Encoder embeds textual targets into a continuous latent space. These embeddings serve as prediction targets and are designed to capture task-relevant semantic information while ignoring unnecessary details.
4. **Y-Decoder:** The Y-Decoder is not used during the primary training stage. During inference, it converts the predicted latent embeddings into human-readable text whenever textual outputs are required.

The VL-JEPA framework is trained in two stages:

1. A pre-training stage using caption datasets to establish robust vision-language alignment, resulting in VL-JEPA_{BASE}.
2. A supervised fine-tuning (SFT) stage that equips the model with visual question answering capabilities, resulting in VL-JEPA_{SFT}.

8.2 Evaluation

The authors evaluate VL-JEPA on a wide range of classification and retrieval benchmarks.

For classification tasks, VL-JEPA_{BASE} achieved top-1 accuracies of:

- SSv2: 19.3
- EK100: 21.8
- EgoExo4D: 33.2
- Kinetics-400: 64.8
- COIN (SR): 47.4
- COIN (TR): 79.4
- CrossTask (SR): 64.5
- CrossTask (TR): 89.6

For retrieval tasks, VL-JEPA_{BASE} achieved Recall@1 scores of:

- MSR-VTT: 40.0
- ActivityNet: 64.9
- DiDeMo: 50.0
- MSVD: 49.0
- YouCook2: 40.0
- PVD-Bench: 83.1
- Dream-1K: 93.3
- VDC-1K: 88.8

After supervised fine-tuning, VL-JEPA_{SFT} demonstrated substantial performance improvements.

Top-1 classification accuracies increased to:

- SSv2: 73.2
- EK100: 44.6
- EgoExo4D: 68.1
- Kinetics-400: 84.8
- COIN (SR): 66.4
- COIN (TR): 90.3
- CrossTask (SR): 79.8

- CrossTask (TR): 96.2

Recall@1 retrieval performance for VL-JEPA_{SFT} was:

- MSR-VTT: 46.2
- ActivityNet: 62.4
- DiDeMo: 52.3
- MSVD: 52.3
- YouCook2: 37.8
- PVD-Bench: 82.6
- Dream-1K: 89.3
- VDC-1K: 87.7

These results demonstrate that VL-JEPA can achieve strong performance across diverse vision-language tasks while maintaining the computational advantages of a non-generative architecture.

9 A Path Towards Autonomous Machine Intelligence Version 0.9.2

The author states in the prologue section that this piece is neither a scholarly paper nor a technical paper, but a position paper to express their vision.

In this position paper, the author proposes an architecture and training paradigm with which to construct autonomous intelligent agents. It combines concepts such as a configurable predictive world model, behaviour driven through intrinsic motivation, and hierarchical joint embedding architectures trained with self-supervised learning.

The main contributions of this paper are:

1. An overall cognitive architecture in which all modules are differentiable and many of them are trainable.
2. JEPA and Hierarchical JEPA: a non-generative architecture for predictive world models that learn a hierarchy of representations.
3. A non-contrastive self-supervised learning paradigm that produces representations that are simultaneously informative and predictable.
4. A way to use H-JEPA as the basis of predictive world models for hierarchical planning under uncertainty.

9.1 Architecture

The H-JEPA architecture consists of 5 differentiable modules:

1. **World Model:** The central component that predicts future states of the world based on the current state and intended actions. It uses a hierarchy of Joint-Embedding Predictive Architectures (JEPAs).
2. **Perception Module:** Acts as an encoder to process raw sensory data (like video or audio) from the environment into useful representations for the rest of the modules.
3. **Actor:** Responsible for generating action plans to accomplish current tasks. It functions at multiple levels of abstraction to convert high-level goals into concrete motor controls.
4. **Critic Module:** Evaluates the predicted outcomes of planned actions, helping the agent find the optimal path under uncertainty.
5. **Configurator:** A meta-controller or gating mechanism that dynamically adjusts the settings of all other modules based on the specific current task.

9.2 Evaluation

The author does not evaluate H-JEPA, a conceptual architecture, on any dataset. Instead, the framework proposes evaluating the model using energy minimization, specifically by measuring the latent space prediction error between predicted and actual future states.

10 LanGWM: Language Grounded World Model

In this paper, Rudra et al., focus on learning language-grounded visual features to enhance the world model learning, a model-based reinforcement learning technique. The authors explore a representation-learning architecture for RL that improves visual features with the help of language.

Rudra et al. proposed the Language Grounded World Model (LanGWM), which enforces language grounding explicitly to improve the generalization of the learned visual features.

In summary, their contributions are two-fold:

- A novel explicitly language grounded visual representation learning method, which outperforms state-of-the-art models on visual control tasks.
- An efficient alternative to visual foundation models for autonomous systems with limited resources.

10.1 Architecture

LanGWM consists of three main components:

1. **Unsupervised Language Grounded Representation Learning:** This uses the object masking-based masked autoencoder.

2. **World Model:** The world model uses a recurrent neural network to output the parameters of a categorical distribution, which is used to sample the future states of the environment.
3. **Controller:** The controller learns the action probability using an actor-critic approach, which maximises the expected reward.

10.2 Evaluation

Rudra et al. evaluated LanGWM’s performance on the PointGoal navigation task using the iGibson 1.0 dataset for out-of-distribution generalisation performance.

They reported Success Rate (SR) and Success weighted by (normalized inverse) Path Length (SPL).

LanGWM achieved:

- Ihlen_0_int environment: 8.3 SR and 0.05 SPL.
- Ihlen_1_int environment: 2.1 SR and 0.01 SPL.
- RS_int environment: 9.9 SR and 0.06 SPL.

The average SR for LanGWM is 6.8, and the average SPL for LanGWM is 0.04.

11 V-JEPA 2.1: Unlocking Dense Features in Video Self-Supervised Learning

In this paper, Lorenzo et al. present V-JEPA 2.1, a family of self-supervised models that learn dense, high-quality representations of visual scenes in both images and videos, with strong global scene understanding.

V-JEPA 2.1 combines four key ingredients:

- A Dense Predictive Loss, a masking-based objective in which all tokens—visible context and masked tokens alike—contribute to the training loss, encouraging explicit spatial and temporal grounding.
- Deep Self-Supervision, which applies the self-supervised objective hierarchically at multiple intermediate encoder layers to improve representation quality.
- Multi-Modal Tokenizers that support unified training over images and videos.
- Effective model and data scaling.

11.1 Architecture

1. Images and videos are processed using 2D or 3D convolutional patch embeddings.
2. 3D RoPE (Rotational Positional Encoding) and modality embeddings are added, and the encoder generates multi-level visual representations.
3. An MLP fuses the multi-level features, reduces dimensionality, and combines them with learnable mask tokens.

4. The predictor processes the combined tokens and generates predictions for the masked regions.
5. Training uses two different losses: (i) an L1 loss on masked-token predictions (the original V-JEPA objective), and (ii) a distance-weighted L1 loss on nearby context tokens, both supervised using the y-encoder multi-level outputs.

11.2 Evaluation

V-JEPA 2.1 achieved state-of-the-art results across multiple benchmarks:

- 7.71 mAP on Ego4D for short-term object-interaction anticipation.
- 40.8 Recall@5 on EPIC-KITCHENS for high-level action anticipation.
- A 20% improvement in real-robot grasping success rate over V-JEPA-2 AC.
- 5.687 ATE on Tartan Drive for robotic navigation.
- 0.307 RMSE on NYUv2 for depth estimation using a linear probe.
- 77.7% accuracy on Something-Something-V2 for global recognition.

These results demonstrate strong performance in visual understanding and world modeling.

12 Diffusion World Model: Future Modeling Beyond Step-by-Step Rollout for Offline Reinforcement Learning

In this paper, Zihan et al. introduce Diffusion World Model (DWM), a conditional diffusion model capable of predicting multistep future states and rewards concurrently.

To evaluate the proposed DWM, the authors use an offline reinforcement learning setting, where policies are learned from a fixed dataset without online interaction. This removes exploration-related effects and enables a clearer assessment of world model quality.

The authors first train a diffusion world model on offline data, then use its generated trajectories to train an actor-critic policy. They also introduce Diffusion-MVE, which estimates target values by simulating future trajectories up to a chosen horizon.

12.1 Architecture

1. The Diffusion World Model (DWM) takes the current state, action, and target Return-to-Go (RTG) as inputs.
2. It predicts future state-reward trajectories rather than future actions, generating an 8-step future trajectory.
3. The core of DWM is a Temporal U-Net composed of residual convolutional blocks for sequence modeling.

4. The diffusion step, RTG, and initial action are encoded into latent embeddings using MLPs.
5. These embeddings are combined and used to condition the diffusion model during trajectory generation.
6. Through a guided diffusion process, the model generates future state and reward sequences that are used for planning and policy learning.

12.2 Results

Zihan et al.’s results confirm that DWM outperforms one-step models, with DWM-based algorithms achieving a 44% performance gain.

They further consider a variant of their approach where the diffusion model is substituted with a Transformer architecture. Their results show that DWM-based algorithms surpass Transformer-based algorithms with a 37.5% performance gain.

13 Value-Guided Action Planning with JEPA World Models

In this paper, Matthieu et al. propose improving planning in JEPA-based world models by shaping the representation space so that the distance between state embeddings reflects the goal-conditioned value function. They introduce a training method that enforces this property and demonstrate that it significantly improves planning performance compared to standard JEPA models on simple control tasks.

13.1 Architecture

1. States are encoded into 512-dimensional latent representations using a convolutional encoder with residual connections.
2. Actions are processed through an identity action encoder without additional transformations.
3. State and action representations are concatenated before being passed to the predictor.
4. An MLP-based predictor learns future state representations in the latent space.
5. The architecture contains a 2.2M-parameter encoder and a 1.3M-parameter predictor.
6. Training is performed on trajectory segments of length 16.
7. The learned representations are shaped so that distances between embeddings approximate the goal-conditioned value function, improving planning performance.

13.2 Evaluation

The authors evaluated their approach in an offline reinforcement learning setting using image-based navigation tasks in the Wall Small (WS), Wall Big (WB), and Maze environments. Planning performance was measured using planning accuracy, and the proposed value-function-guided JEPA representations were compared against contrastive, regressive, and standard prediction-based JEPA approaches.

13.3 Results

Matthieu et al. report that the VF quasi approach achieved the highest planning accuracies across all environments, reaching 0.71 in WS, 0.96 in WB, and 0.63 in Maze. These results outperformed the standard prediction-based JEPA approach, which achieved accuracies of 0.55, 0.89, and 0.54 respectively, highlighting the effectiveness of value-function-guided representations for planning.

14 Diffusion for World Modeling: Visual Details Matter in Atari

In this paper, Eloi et al. introduce DIAMOND (Diffusion As a Model of eNvironment Dreams), a reinforcement learning agent trained in a diffusion world model. They analyze the key design choices required to make diffusion stable for world modeling and demonstrate how improved visual details can lead to improved agent performance.

14.1 Architecture

- The architecture is built around a diffusion-based world model called DIAMOND.
- The world model predicts future observations directly in pixel space using a diffusion process rather than discrete latent tokens.
- The diffusion model is conditioned on previous observations and agent actions to generate future environment states.
- An RL agent is trained entirely inside the generated environment produced by the world model.
- The authors employ the EDM (Elucidated Diffusion Model) framework to improve generation quality and stability.
- The generated trajectories are used to train policies without requiring additional interaction with the real environment.

14.2 Evaluation and Results

Eloi et al. evaluated DIAMOND on the Atari 100k benchmark, where agents are limited to 100,000 interactions with the environment. They compared DIAMOND against several state-of-the-art world-model-based reinforcement learning approaches, including

SimPLe, TWM, IRIS, DreamerV3, and STORM, using Human Normalized Score (HNS) and Interquartile Mean (IQM) as evaluation metrics.

The authors report that DIAMOND achieved a mean Human Normalized Score (HNS) of 1.46 and an Interquartile Mean (IQM) of 0.64, establishing a new state-of-the-art result among agents trained entirely within a world model. DIAMOND also outperformed human players on 11 Atari games and demonstrated particularly strong performance on visually demanding games such as Asterix, Breakout, and Road Runner, highlighting the importance of preserving visual details during world modeling.

15 A Generalization Theory for JEPA-Based World Models

In this paper, Jingyi et al. develop the first generalization theory for JEPA-based world models. They formulate JEPA pretraining as a conditional spectral graph learning problem and show that the JEPA objective is equivalent to a low-rank factorization of an action-conditioned co-occurrence matrix.

The main contributions of this paper are:

- The authors introduce a spectral graph-based theoretical framework for JEPA-based world models using an action-conditioned state co-occurrence matrix.
- They show that JEPA training can be interpreted as factorizing this action-conditioned co-occurrence matrix and derive a generalization error bound for JEPA-based world models.
- Their analysis reveals a trade-off between approximation error and sample error with respect to latent dimensionality, providing theoretical insights into the strengths of latent-level and input-level predictive models.

15.1 Approach

Rather than proposing a new world model architecture, Jingyi et al. develop the first theoretical framework for JEPA-based world models. They formulate JEPA learning as a conditional spectral graph problem using an action-conditioned co-occurrence matrix, show that JEPA training is equivalent to low-rank matrix factorization, and derive generalization bounds linking pretraining error to downstream planning regret. Their analysis also reveals a trade-off between latent dimensionality, approximation error, and sample efficiency, providing theoretical insights into the strengths and limitations of latent predictive world models.

15.2 Evaluation

Jingyi et al. evaluated their theory using synthetic control environments with different noise levels and planning horizons. By comparing latent-level JEPA models with input-level predictive models, they showed that latent representations are more robust to noise and provide more stable long-horizon planning performance. Their experiments also validated the theoretical connection between JEPA pretraining error and downstream planning regret, as well as the trade-off between approximation error and sample error predicted by their analysis.

16 LLM-JEPA: Large Language Models Meet Joint Embedding Predictive Architectures

In this paper, Hai et al. propose a first step toward developing LLM-JEPA, a JEPA-based framework for large language models that can be applied to both finetuning and pretraining. The authors demonstrate that LLM-JEPA consistently outperforms standard LLM training objectives across multiple models while remaining robust to overfitting.

The main contributions of this paper are:

- **Novel JEPA-based training objective:** The authors introduce the first JEPA-based training objective for LLMs that operates in embedding space while preserving the generative capabilities of autoregressive language models.
- **Improved performance:** They empirically demonstrate that LLM-JEPA improves performance over standard LLM training objectives in both finetuning and pretraining settings.
- **Extensive empirical validation:** The approach is evaluated across multiple model families (Llama, Gemma, OpenELM, and OLMo), datasets (NL-RX, GSM8K, Spider, and Rotten Tomatoes), and model sizes.

16.1 Architecture

The construction of LLM-JEPA is based on two principles: preserving the generative capabilities of LLMs and improving their abstraction capabilities through joint embedding prediction. The architecture consists of the following components:

1. A standard autoregressive LLM serves as the backbone model.
2. An encoder extracts latent embeddings from text sequences using the final hidden representation.
3. A predictor network maps the input text embedding to the embedding of a target text.
4. The target text is encoded to provide the prediction target in the latent space.
5. Training combines the conventional next-token prediction objective with a JEPA embedding prediction objective.
6. The model learns both generative capabilities and abstract semantic representations through the joint training objective.

16.2 Results

Hai et al. evaluated LLM-JEPA on NL-RX (Natural Language to Regular Expressions), GSM8K (Mathematical Reasoning), Spider (Text-to-SQL Generation), and Rotten Tomatoes (Sentiment Classification) using Llama, Gemma, OpenELM, and OLMo models. The authors report that LLM-JEPA consistently outperformed standard next-token prediction training across all datasets and model families. On the NL-RX-SYNTH

benchmark, Llama-3.2-1B-Instruct improved from 57.29% accuracy with standard fine-tuning to 71.46% accuracy with LLM-JEPA. Similar gains were observed across Gemma 2, OpenELM, and OLMo models, demonstrating the benefits of combining JEPA objectives with conventional language modeling.

17 GOT-JEPA: Generic Object Tracking with Model Adaptation and Occlusion Handling using Joint-Embedding Predictive Architecture

In this paper, Shih-Fang et al. propose GOT-JEPA, a model-predictive pre-training framework that extends JEPA from predicting image features to predicting tracking models.

The main contributions of this paper are:

- GOT-JEPA, a JEPA-based framework that learns to predict tracking models rather than image features.
- A teacher-student predictive learning strategy that improves robustness to occlusions, distractors, and appearance variations.
- OccuSolver, a module for fine-grained object-aware visibility estimation and occlusion pattern modeling.
- Strong improvements in tracking robustness and generalization across multiple tracking benchmarks.

17.1 Architecture

The GOT-JEPA architecture consists of two main components:

17.1.1 GOT-JEPA Framework

1. Historical frames and previous tracking results provide contextual information about the target.
2. A Teacher Predictor processes clean frames and generates pseudo tracking models.
3. A Student Predictor receives corrupted frames and learns to predict the teacher’s tracking models.
4. The teacher-student JEPA objective improves robustness to appearance changes, distractors, and noisy observations.

17.1.2 OccuSolver

5. OccuSolver performs fine-grained visibility estimation using point-based object tracking.
6. It models occlusion patterns and refines object visibility information.
7. The refined visibility cues are used to improve tracking performance under partial and full occlusions.

17.2 Evaluation

Shih-Fang et al. evaluated GOT-JEPA on seven generic object tracking benchmarks to measure tracking accuracy, robustness, generalization to unseen targets, and performance under adverse visibility conditions. The evaluation included challenging scenarios involving occlusions, distractors, deformation, and background clutter.

17.3 Results

Shih-Fang et al. report that GOT-JEPA achieved a 63.7% success rate on AViT, surpassing PiVOT (62.2%), and an average overlap of 79.6% on GOT-10k. The results demonstrate strong tracking accuracy, generalization, and robustness to occlusions.

18 Text-Conditional JEPA for Learning Semantically Rich Visual Representations

In this work, Chen et al. propose Text-Conditional JEPA (TC-JEPA), which uses image captions to reduce prediction uncertainty. TC-JEPA combines the predictive learning framework of JEPAs with text conditioning, an under-explored direction in visual representation learning. By incorporating textual guidance, the authors transform I-JEPA into a text-conditional representation learner, where patch representations become more predictable, language-aligned, and semantically meaningful.

The main contributions of this paper are:

- They propose TC-JEPA to improve I-JEPA through fine-grained text conditioning, resulting in semantically rich visual representations.
- TC-JEPA improves downstream performance, training stability, and scaling properties.
- TC-JEPA introduces a fine-grained vision-language pretraining paradigm based solely on feature prediction and outperforms contrastive methods on diverse fine-grained tasks.

18.1 Architecture

The architecture consists of four main components:

1. **Image Encoder (ViT)** – extracts visual representations from visible image patches.
2. **Target Encoder** – generates target representations for masked image patches.
3. **Predictor Network** – predicts representations of masked patches from context features.
4. **Text Conditioner** – processes image captions and injects textual information into visual representations through cross-attention.

18.2 Evaluation

Chen et al. evaluated TC-JEPA on visual and vision-language tasks using ImageNet-1K, CC12M, and YFCC15M. The model was compared against I-JEPA, MAE, data2vec, CLIP, and SigLIP using image classification, semantic segmentation, and vision-language benchmarks.

18.3 Results

Chen et al. report that TC-JEPA consistently outperformed I-JEPA across different model scales. On ImageNet-1K linear probing, TC-JEPA achieved 80.4% Top-1 accuracy with ViT-H/14, compared to 79.3% for I-JEPA. The model also surpassed MAE and data2vec while demonstrating stronger semantic understanding and improved training efficiency.

19 Semantic Tube Prediction: Beating LLM Data Efficiency with JEPA

In this paper, Hai et al. propose a novel Semantic Tube Prediction (STP) task, a JEPA-style regularizer that confines hidden-state trajectories to a tubular neighbourhood of a geodesic. STP generalizes JEPA to language without requiring explicit multi-view augmentation.

The main contributions of this paper are:

- The authors introduce the Geodesic Hypothesis, which assumes that token sequences follow smooth semantic trajectories that are locally linear in latent space.
- They propose Semantic Tube Prediction (STP), a JEPA-style regularizer that constrains hidden-state trajectories to remain within a semantic tube around the underlying geodesic.
- They generalize JEPA to language modeling without requiring explicit multi-view augmentations.
- They demonstrate substantially improved data efficiency, challenging the data-efficiency limits suggested by Chinchilla-style scaling laws.

19.1 Architecture

The architecture consists of four main components:

1. **Autoregressive LLM Backbone** – generates hidden-state representations from token sequences.
2. **Hidden-State Trajectory Modeling** – treats successive hidden states as points on a semantic manifold.
3. **Semantic Tube Prediction (STP) Module** – constrains hidden-state trajectories to remain close to a locally linear geodesic path.
4. **Joint Training Objective** – combines standard next-token prediction with the STP regularization loss.

19.2 Evaluation

Hai et al. evaluated Semantic Tube Prediction (STP) on the NL-RX-SYNTH benchmark and additional reasoning and language understanding tasks across multiple model families and model sizes. The experiments focused on measuring data efficiency and comparing STP against standard next-token prediction training and LLM-JEPA.

19.3 Results

Hai et al. report that STP consistently outperformed standard fine-tuning and LLM-JEPA across datasets, model families, and model sizes. Most notably, STP matched baseline accuracy on NL-RX-SYNTH using $16\times$ less training data, demonstrating substantially improved data efficiency and challenging conventional scaling-law assumptions.

20 A-JEPA: Joint-Embedding Predictive Architecture Can Listen

In this paper, Zhengcong et al. introduce Audio-based Joint-Embedding Predictive Architecture (A-JEPA), a simple extension of JEPA for self-supervised learning from audio spectrograms.

Following the design of I-JEPA, A-JEPA uses a context encoder with a curriculum masking strategy to encode visible audio spectrogram patches. The model then predicts the representations of selected masked regions. Target representations for these regions are generated by a target encoder, implemented as an exponential moving average (EMA) version of the context encoder, which processes the entire spectrogram.

The main contributions of this paper are:

- The authors introduce A-JEPA, the first JEPA-based framework for self-supervised audio representation learning.
- They propose curriculum masking and regularized patch masking to better capture audio patterns and improve fine-tuning.
- A-JEPA demonstrates strong scalability and outperforms AudioMAE on AS-2M classification and several downstream audio tasks.

20.1 Architecture

The A-JEPA architecture consists of four main components:

1. **Context Encoder** – encodes visible spectrogram patches.
2. **Target Encoder** – generates target representations from the full spectrogram.
3. **Predictor Network** – predicts the representations of masked regions.
4. **Curriculum Masking Mechanism** – progressively masks time-frequency regions during training.

During pretraining, A-JEPA learns by predicting latent representations of masked regions rather than reconstructing audio pixels.

20.2 Evaluation

The authors evaluated A-JEPA on AudioSet-2M (AS-2M) and several downstream audio tasks, including audio classification, audio tagging, and sound event recognition. They compared A-JEPA against self-supervised audio representation learning methods, particularly AudioMAE, to evaluate scalability and representation quality.

20.3 Results

The authors report that A-JEPA consistently outperformed reconstruction-based methods such as AudioMAE on AudioSet-2M classification and multiple downstream audio tasks. The results demonstrate that predictive representation learning is more effective than pixel-level reconstruction for learning audio representations. The experiments also show that A-JEPA scales effectively with larger models and datasets while maintaining strong transfer performance across diverse audio benchmarks.